

MALYE KARMAKULY, Russian Federation

WMO: 207440

Lat: 72.373N

Lon: 52.717E

Elev: 62

StdP: 14.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.7	-13.8	-24.2	1.4	-16.0	-20.8	1.8	-13.1	74.4	-8.8	68.6	-10.7	19.4	100	1.107

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	61.6	54.9	57.3	52.1	53.7	49.6	55.7	60.2	52.6	56.5	49.9	53.1	18.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
53.1	60.1	58.7	50.0	53.5	55.0	47.5	48.8	51.9	23.7	60.2	21.8	56.9	20.3	53.3	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 58.7	49.8	42.4	-17.2	65.2	4.5	6.4	-20.4	69.8	-23.0	73.5	-25.5	77.1	-28.7	81.8		
(5)			-17.1	57.9	6.0	3.8	-21.5	60.7	-25.0	63.0	-28.3	65.1	-32.7	67.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	9.4	8.2	11.2	17.0	26.2	37.4	45.9	44.5	39.7	29.8	20.6	15.3
	DBStd	16.10	12.89	13.30	11.42	8.94	7.18	6.06	6.86	4.65	4.79	7.60	9.41	11.62
	HDD50	8984	1258	1170	1203	990	738	383	168	181	311	625	882	1075
	HDD65	14400	1723	1590	1668	1440	1203	828	592	634	760	1090	1332	1540
	CDD50	60	0	0	0	0	0	5	42	12	1	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.3	20.7	21.7	19.6	18.3	15.5	15.3	13.0	13.9	14.5	16.8	18.7	20.1
Precipitation	PrecAvg	13.5	1.1	0.8	0.8	0.7	0.7	0.9	1.7	1.5	1.9	1.3	0.9	1.1
	PrecMax	18.9	3.4	3.7	2.8	2.5	1.6	1.9	4.8	4.3	5.0	3.4	2.0	4.1
	PrecMin	9.4	0.2	0.1	0.2	0.2	0.2	0.1	0.0	0.0	0.1	0.4	0.1	0.2
	PrecStd	2.6	0.9	0.7	0.6	0.5	0.4	0.5	1.1	1.1	1.0	0.7	0.5	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	32.3	33.0	35.5	48.5	61.0	69.3	61.4	54.0	45.4	35.8	34.3
		MCWB	32.4	31.5	31.6	33.5	42.2	52.9	59.8	55.0	49.8	43.3	34.4	33.0
	2%	DB	31.7	30.8	31.5	32.9	40.7	55.3	64.3	56.9	50.7	42.4	34.3	33.0
		MCWB	30.8	29.9	30.4	31.7	36.8	48.9	56.6	52.0	47.8	40.9	33.1	32.1
	5%	DB	29.9	29.2	30.0	31.4	36.8	50.3	60.5	54.0	48.4	40.7	33.2	31.9
		MCWB	28.8	28.3	28.8	30.4	34.5	44.6	54.6	50.5	46.3	39.5	32.0	31.1
	10%	DB	27.4	26.7	28.1	29.9	34.8	46.1	56.3	51.4	46.5	38.8	32.0	30.6
		MCWB	26.3	25.6	26.9	28.7	33.0	42.0	51.6	48.2	44.7	37.6	30.9	29.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.8	31.7	32.1	33.9	42.6	54.2	60.8	56.0	50.7	43.6	34.8	33.6
		MCDB	33.0	31.8	32.6	35.1	48.5	59.1	67.3	59.6	53.2	44.9	35.3	33.8
	2%	WB	30.9	30.2	30.5	32.1	37.2	49.6	57.9	53.0	48.4	41.3	33.5	32.3
		MCDB	31.5	30.7	31.3	32.7	39.5	55.4	63.4	55.9	50.3	42.2	34.1	32.7
	5%	WB	28.9	28.3	28.9	30.6	34.9	45.0	55.1	50.7	46.4	39.6	32.3	31.1
		MCDB	29.8	29.1	29.9	31.3	36.5	49.2	59.8	53.4	47.9	40.6	33.1	31.8
	10%	WB	26.3	25.7	26.9	28.8	33.2	42.2	51.8	48.6	44.8	37.8	31.0	29.6
		MCDB	27.4	26.6	28.1	29.8	34.6	45.6	55.9	51.1	46.4	38.7	32.0	30.5
Mean Daily Temperature Range	5% DB	MDBR	8.8	8.8	8.8	9.0	6.6	5.6	7.1	6.0	5.2	5.4	7.1	8.5
		MCDBR	8.2	8.6	8.1	8.1	8.0	11.7	12.8	10.2	7.1	4.6	5.3	7.2
		MCWBR	8.5	8.8	8.2	8.0	5.8	8.1	8.1	7.6	5.7	4.7	5.6	7.4
	5% WB	MCDBR	8.1	8.7	8.0	8.1	7.4	10.8	12.7	9.3	6.3	4.5	5.4	7.1
		MCWBR	8.5	8.9	8.2	8.2	5.6	8.0	8.5	7.4	5.5	4.6	5.8	7.4
Clear-Sky Solar Irradiance	taub	N/A	0.229	0.287	0.313	0.329	0.325	0.348	0.330	0.299	N/A	N/A	N/A	N/A
	taud	N/A	1.928	2.076	2.136	2.207	2.361	2.410	2.452	2.427	N/A	N/A	N/A	N/A
	Ebn at Noon	0	109	207	250	264	271	259	248	216	N/A	0	0	0
	Edh at Noon	0	10	23	32	35	31	29	24	17	N/A	0	0	0
All-Sky Solar Radiation	RadAvg	0	53	443	1144	1731	1765	1472	919	452	114	0	0	0
	RadStd	0	7	56	84	103	179	136	97	36	15	0	0	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+2.83	+4.68	+4.99	+3.15	+3.06	+3.28	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page